



Two-way coupling dynamics of CH₄ adsorption and coal matrix deformation: Insights from hybrid GCMC/MD simulations

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ABSTRACT

CH₄ adsorption can deform coal microporous structures, subsequently altering adsorption isotherms. To unravel this intricate interplay at a microscopic level, we use a hybrid Grand Canonical Monte Carlo/Molecular Dynamics (GCMC/MD) simulation at 313.15 K and pressures up to 500 bar on five independent amorphous coal matrix models. Our results reveal that CH₄ adsorption increases pore volume and porosity primarily by generating additional pores of similar sizes to those present in coal matrices, thereby maintaining a consistent average pore size across different pressures. The volumetric strain has a linear correlation with CH₄ loading, with volumetric swelling amount approximating the expansion of CH₄-occupiable pore volume, but less than He-occupiable pore volume which results in increased matrix skeletal density. Local radial density distribution of carbon atoms indicates that the immediate environment around carbon atoms remains unchanged. In a flexible matrix, the energy released during CH₄ adsorption is partially absorbed by matrix deformation, resulting in a lower isothermic heat of adsorption compared to a rigid matrix, which suggests easier desorption. This study provides new insights into the mutual relationship between CH₄ adsorption and coal matrix deformation, shedding lights on the complex interactions of various hydrocarbons with geomaterials.

1. Introduction

As an unconventional natural gas resource, coalbed methane (CBM) provides a clean alternative to traditional fossil fuels, enhancing energy security while significantly reducing greenhouse gas emissions [1–3]. During CBM development, CH₄ adsorption/desorption can lead to coal deformation, which significantly influences its permeability and consequently impacts gas recovery efficiency [4–7]. A large number of theoretical and experimental studies have been conducted to unravel the relationship between coal matrix deformation and its permeability [8–13]. These studies [14,15] generally indicate that coal swelling primarily arises from gas adsorption within micropores (<2 nm). CH₄ adsorption-induced deformation can alter the internal microporous structures of coal matrices, which in turn can influence gas adsorption. Whereas, prevailing experimental studies predominantly focus on macroscopic strain [16–18], while generally ignoring microscopic phenomena taking place within micropores. In addition, a constant pore

volume pre-determined by helium expansion is often used to assess CH₄ adsorption in experimental measurements [19,20], which becomes inapplicable as coal microporous structures evolve due to CH₄ adsorption. Therefore, understanding the interplay between CH₄ adsorption and nanoporous coal deformation is imperative to the accurate assessment of gas recovery from coal and optimization of CBM development.

Nowadays, molecular simulations including molecular dynamics (MD) simulations [21] and Monte Carlo (MC) simulations [22,23] have become a powerful tool to unravel the complex adsorption-induced deformation behaviors of microporous solid. The intricate process which involves gas loading and solid structural changes often warrants the integration of Grand Canonical Monte Carlo (GCMC) simulations with MD simulations [24]. The GCMC moves can effectively circumvent diffusion-limited processes commonly encountered in MD simulations, thereby expediting the equilibration processes. The combination of GCMC and MD techniques has facilitated in-depth exploration of adsorption-induced deformation behaviors in various flexible

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microporous materials such as zeolites [25,26], polymers [27,28], and metal–organic frameworks (MOFs) [29,30].

Brochard *et al.* pioneered the quantification of coal swelling arising from the replacement of CH₄ by CO₂ under field conditions using a micro-poromechanical model through molecular simulations [31]. This model has since been extensively adopted by later studies [32–37], in which the correlations between coal/shale matrix volumetric strain and adsorption pressure are obtained based on a constant coupling coefficient ($C_{\text{methane}} = 6.05 \pm 7\%$; $C_{\text{carbon dioxide}} = 7.60 \pm 20\%$) and bulk modulus ($K=2.65$ GPa) from Brochard *et al.* [31]. However, two notable deficiencies are present in the aforementioned studies: 1) the coupling coefficient in the poromechanical model requires explicit determination for different systems due to the adsorbate-adsorbent pair characteristics and potential temperature dependence [38]. Moreover, the assumption of constant mechanical properties for porous media during adsorption disregards porosity changes which are typically linked to mechanical properties (e.g. bulk modulus) and vary with matrix deformation; 2) The sequential “first-GCMC-then-MD” approach (one-way coupling) fails to account for the dynamic interplay between adsorption and deformation, in which gas adsorption influences the mechanical properties of adsorbent and *vice versa*. On the other hand, Ravikovitch and Neimark developed a molecular-level approach to correlate adsorption with strain measurements, enabling the calculation of mechanical properties in nanoporous materials with spheroidal pores [39]. Neimark and his coworkers later extended this work by predicting deformation effects during pure and multicomponent gas adsorption based on adsorption stress, integrating data from experiments and simulations (SAFT-DFT, GCMC) data [40–42]. However, their approach is not a “direct” simulation of adsorption-induced deformation which relies on assumptions including linear elastic deformation and the neglect of material anisotropy.

To simulate two-way adsorption-deformation coupling directly, a robust hybrid/alternating strategy that combines GCMC and MD is essential. This approach enables the so-called osmotic statistical ensemble in which chemical potential of adsorbate, system stress, and temperature are fixed. This hybrid GCMC/MD method has been widely used to study kerogen deformation due to gas adsorption. Wu *et al.* [43] developed a non-linear adsorption-strain model to predict adsorption-induced kerogen deformation. A number of studies [44–47] reported that using rigid kerogen model may lead to erroneous gas adsorption estimation. On the other hand, Yu *et al.* [48] reported a unified correlation between volumetric strain and absolute adsorption amount across various adsorbates and introduced a theoretical framework for computing excess adsorption isotherms by considering swelling effects. Interestingly, Potier *et al.* [38] reported that their constructed 3D-connected mature kerogen model exhibits viscoelasticity, resulting from argon adsorption at 243 K after hundreds of nanoseconds of simulations. Both matrix internal rearrangements and volumetric swelling contribute to the gas loading increase compared to rigid structure. While these studies provided important insights into two-way coupling between gas adsorption and deformation, some questions still remain unanswered. For example, how does adsorption affect micropore structures and solid spatial arrangement? How do dynamic changes in pore space influence gas adsorption behaviors? Furthermore, the amorphous structures of coal and kerogen introduce significant structural randomness, which often challenges the reproducibility of reported results.

Therefore, in this work, we employ a hybrid GCMC/MD approach in the isothermal-isostress ($N\sigma T$) ensemble to investigate the adsorption-induced deformations of five distinct coal matrices. These simulations can effectively study gas adsorption behaviors by explicitly accounting for coal mechanical deformations under external (or confining) stress. This external stress typically mirrors the constraint imposed by the adsorbing pressure, known as “drained” or “unjacketed” conditions, wherein the external stress equals the fluid pressure, allowing the observation of maximum deformation [49]. We analyze the relationship between CH₄ adsorption amount and coal matrix volumetric strain, the

effects of adsorption on pore structure (e.g. pore volume, porosity, pore size distribution, average pore size) evolution and solid structural properties, and the influence of adsorption-induced deformation on adsorption behavior in coal matrices. The insights gained from this study enhance fundamental understandings about the dynamic CH₄ adsorption process in coal matrices at molecular scale, which are important to CBM development and its optimization.

2. Molecular model and simulations

2.1. Microporous coal matrix model generation

The type III-A kerogen unit (C₂₃₃H₂₀₄O₂₇N₄, molecular weight 3492.12 g/mol), developed by Ungerer *et al.* [50], is utilized as a representative macromolecule of bituminous coals. Kelemen *et al.* [51] reported that the III-A model pertains to low maturity Cretaceous or Tertiary coals, with aromatic carbon ring sizes in accordance with bituminous coals at comparable aromatic carbon levels. In addition, the element ratio of the III-A model (H/C=0.876, O/C=0.116) agrees well with the Wiser’s coal model (C₁₉₁H₁₆₉O₂₁N₃S₃, H/C=0.885, O/C=0.110) [52], within the range of bituminous coal on the Van Krevelen diagram [53,54]. The consistent valence force field (CVFF) [55,56] and TraPPE-UA [57] models are utilized to simulate coal macromolecules and CH₄ molecules, respectively. A cutoff distance of 1.4 nm is applied for pairwise 12–6 Lennard-Jones (LJ) and Coulombic interactions. Tail corrections are employed for long-range LJ interactions and long-range Coulombic interactions are computed using the Ewald summation method with an accuracy set as 1×10^{-4} . Pairwise energy and force contributions between atoms bonded directly or via one or two intermediate bonds are excluded. The Lorentz-Berthelot arithmetic mixing rule is applied to determine the LJ coefficients (σ_{LJ} and ϵ_{LJ}) for interactions between unlike atoms. Periodic boundary conditions are implemented in all dimensions.

To generate three-dimensional (3D) microporous coal matrix models, a staged cooling technique is employed as shown in Fig. 1, involving multiple iterations of MD simulations under elevated temperatures and pressures to allow the coal matrix to thoroughly explore its free energy landscape. Initially, 15 III-A units are randomly placed in a cubic cell of 10 nm × 10 nm × 10 nm without overlap using Packmol [58], resulting in a very low initial average density (~ 0.1 g/cm³). Using LAMMPS [59] with a time step of 0.5 fs, several conformations are generated through an NVT simulation at 1000 K and low density (~ 0.1 g/cm³) with various solid structures. Each conformation is then relaxed in successive steps of NPT simulations from 1000 K down to 298.15 K. Temperature control is achieved using a three-chain Nosé-Hoover method with a damping parameter of 100 fs [60]. An anisotropic Nosé-Hoover barostat with a damping parameter of 1000 fs is used to independently regulate the three diagonal components of stress tensor, ensuring perfectly isotropic pressure within the anisotropic system [61]. The complete relaxation process is summarized in Table 1. Relaxation results from step 10 and 12 are used for hybrid GCMC/MD simulations and model validations, respectively.

To ensure the validity of our relaxed coal models, we compare various structural properties to available experimental data, including density, micropore volume, pore size distribution, and bulk modulus, as well as CH₄ adsorption isotherms. The detailed validations can be referred to Supporting Information (SI, Figs. S1–S3).

2.2. Simulation methods

The simulation of CH₄ adsorption-induced coal matrix deformation employs a hybrid GCMC/MD approach within LAMMPS. In our GCMC simulations for the coal-CH₄ system, we employ the Metropolis algorithm to accept or reject CH₄ molecule insertion and deletion moves. This algorithm evaluates the internal energy change of each move, considering interactions with the heterogeneous coal matrix and other

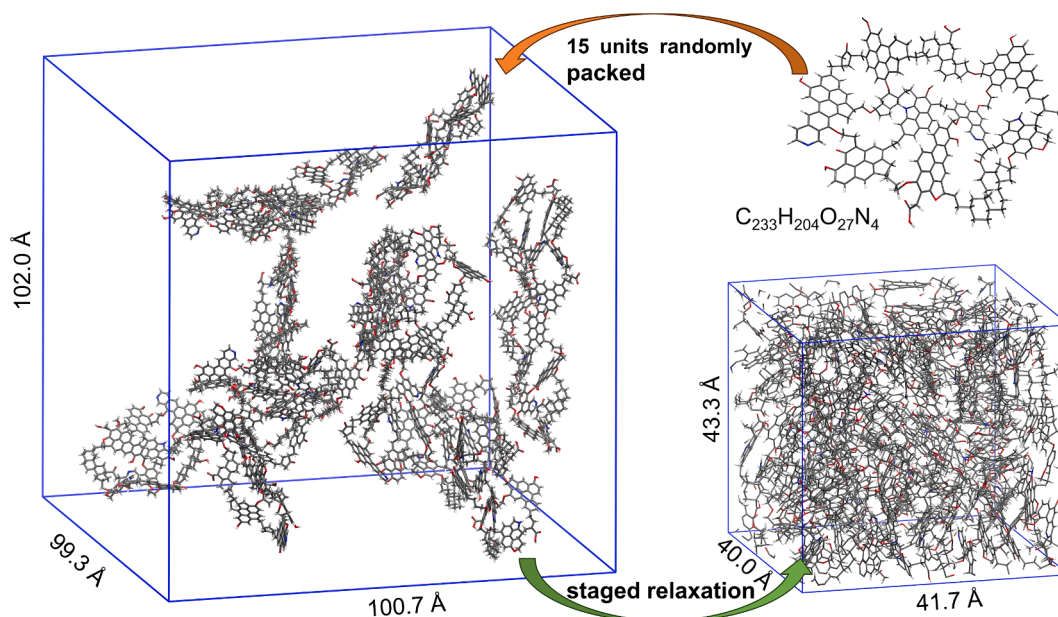


Fig. 1. Schematic representation of generating one representative coal matrix model. The image is generated by using VMD [62].

Table 1

MD relaxation procedure to create coal matrices.

Step	Ensemble	Temperature (K)	Pressure (bar)	Simulation time (ns)
1	NVT	1000	—	1
2	NPT	1000	100	1
3	NPT	1000 ~ 700	100	5
4	NPT	700	100	1
5	NPT	700 ~ 500	100	2
6	NPT	500	100	1
7	NPT	500 ~ 313.15	100	2
8	NPT	313.15	100	1
9	NPT	313.15	100 ~ 0	2
10	NPT	313.15	0	20
11	NPT	313.15 ~ 298.15	0 ~ 1	2
12	NPT	298.15	1	20

CH₄ molecules. Moves are accepted based on the Metropolis criterion, which allows for both energy-lowering moves and, with a certain probability, energy-increasing moves, ensuring the simulation accurately captures the adsorption behavior of CH₄ in coal under constant chemical potential (μ), volume (V) and temperature (T). In addition, it allows dynamic movements of gas molecules along with the local relaxation of coal matrix using MD in *NPT* ensemble, where pressure (P) is controlled anisotropically. Typically, the input parameter μ can be substituted by fugacity from various equations of state (EOS) models [38,48,63,64]. However, the properties (such as P) from molecular models often deviate from those predicted by EOS models [65] (see Fig. S4). Therefore, we obtain the dependence of P on μ by conducting additional hybrid GCMC/MD (*NVT*) simulations of bulk gas phase as shown in Fig. S5. The obtained P – μ pair is used as an input for hybrid GCMC/MD simulations. We perform 30,000 cycles for the hybrid GCMC/MD simulations, with each cycle consisting of 1,000 GCMC insertion/deletion attempts with equal probability for CH₄ molecules followed by 1,000 *NPT* MD steps for CH₄ molecules and matrix. The initial 15,000 cycles are for equilibration, and the remaining 15,000 cycles are for sampling. We also run the hybrid GCMC/MD simulations with rigid coal matrices for comparison. The efficiency of phase space sampling is confirmed by monitoring the average matrix volume and number of CH₄ molecules throughout the simulations (see Fig. S6). Notably, this hybrid method ensures chemical, thermal and mechanical equilibria between the system and the imaginary bulk CH₄ reservoir,

mimicking experimental adsorption conditions where coal samples are immersed in a bulk gas reservoir with zero effective stress (“drained” or “unjacketed” conditions) on the solid.

2.3. Pore structure characterization

We use PoreBlazer v4.0 (PB v4.0) [66] to calculate porosity, surface area, pore volume, and pore size distribution (PSD) of a coal matrix. Compared to other software, such as Zeo++ [67], RASPA [68], PorosityPlus [69], PB v4.0 is particularly suitable for analyzing the structure of disordered materials. Additionally, it examines properties affected by intermolecular interactions, extending its analysis beyond purely geometric features. While PB v4.0 can compute accessible surface area using a probe particle rolling over the atomic surface of the structure, we note that in microporous solids, the concept of pore surface area becomes ambiguous as gas adsorption primarily takes place via micropore filling [70,71]. Therefore, the surface area data may not be directly compared with experimental values based on BET theory [72]. In this work, pore volume refers to the probe-occupiable volume, defined as the space enclosed by the Connolly surface. The spherical probes have diameters of 0.373 nm for CH₄ and 0.258 nm for He. We note that the geometric parameters reported are calculated without considering pore accessibility. This approach is consistent with the GCMC algorithm, which does not distinguish between accessible and inaccessible ones [73].

While physical adsorption experiments using N₂/CO₂/Ar have become standard for characterizing porous materials [74,75], accurately determining PSD still remains challenging [66]. Firstly, resolving the adsorption integral equation (AIE) is difficult due to its ill-posed and ill-conditioned nature, often leading to multiple or no solutions for PSD. Additionally, PSD is highly sensitive to minor variations in input adsorption isotherms. Secondly, although modern characterization methods use DFT to model isotherms for simple geometric pores, its applicability to the complex network of irregular pores in coal matrix is uncertain [72]. Lastly, current approaches neglect adsorption-induced deformation of porous framework, resulting in PSD outputs limited to conditions of low pressures (e.g. N₂ P/P_0 : 0.001 ~ 0.995, CO₂ P/P_0 : 0.0001 ~ 0.0289) and low temperatures (e.g. N₂: 77 K, CO₂: 273.15 K). As a result, the evolution of pore space during adsorption-induced coal matrix deformation remains unexplored. However, using a single sorting algorithm in PB v4.0, geometric PSDs at any stages of matrix deformation can be fully examined based on the configuration of solid atoms. To

ensure the statistical accuracy of the results, error bars for all pore structure properties are computed as the standard error of the mean from five independent configurations, or from block averages in the production run for other quantities (e.g. gas loading and volumetric strain).

To ensure rigorous statistical analysis, we select five distinct coal matrix models, each derived from the same relaxation process but exhibiting unique final pore structures due to inherent variability, as shown in Table 2 and Fig. S2.

3. Results and discussion

3.1. Relationship between volumetric swelling and CH₄ adsorption

We plot CH₄ adsorption isotherms within both flexible and rigid matrices of five samples in Fig. 2(a), which are fitted with the Tóth isothermal model [76],

$$n(P) = n_{\max} \frac{kP}{[1 + (kP)^t]^{1/t}} \quad (1)$$

where $n(P)$ is the CH₄ loading at various equilibrium P , $n_{\max}$ is the theoretical maximum adsorbed amount, k is the Tóth equilibrium constant reflecting the adsorbate and adsorbent affinity, with a higher k value suggesting stronger binding, and t represents the heterogeneity of the adsorption sites, with lower values of t (<1) indicating more heterogeneity while $t = 1$ implies homogeneity (Langmuir adsorption). The fitted parameters are presented in Table 3. Note that we select the Toth model for its ability to account for the heterogeneity of adsorption sites, providing an excellent agreement with our data. In contrast, the Langmuir and SBET models assume an energetically homogeneous surface [72], which fails to capture the pore size and surface heterogeneity in coal. The SDR model, while based on a pore filling mechanism, assumes constant adsorbed phase density or volume, making it less suitable for deformable matrix [72].

All adsorption isotherms show excellent fitting quality ($R^2 > 0.99$) with the Tóth isothermal model. In line with Ho et al. [44] and Potier et al. [38], our results confirm that ignoring matrix flexibility significantly underestimates CH₄ adsorption capacity. As indicated by the fitted lines, Sample #2 with the smallest initial pore volume exhibits the largest increase in CH₄ loading (129.01 % at 500 bar) when matrix flexibility is considered, while Sample #4 with the largest initial pore volume shows the smallest increase (61.24 % at 500 bar). This trend also

Table 2
Pore properties of five different coal matrix models.

	Sample #1	Sample #2	Sample #3	Sample #4	Sample #5
CH ₄ -occupiable pore volume (cm ³ /g)	0.072	0.040	0.063	0.073	0.051
He-occupiable pore volume (cm ³ /g)	0.126	0.097	0.114	0.134	0.107
Geometric pore volume (cm ³ /g)	0.281	0.258	0.279	0.290	0.267
Porosity (% CH ₄ as probe)	8.650	4.933	7.594	8.708	6.188
Pore limiting diameter, PLD (nm)	0.192	0.193	0.169	0.232	0.187
Largest cavity diameter, LCD (nm)	0.819	0.546	0.953	0.927	0.749

Note: The geometric pore volume is the region of the system not occupied by atoms. PLD refers to the largest probe that can traverse the simulation cell in at least one dimension via a diffusive pathway, while LCD denotes the largest pore in the structure [66].

holds for $n_{\max}$. Compared to rigid matrices, the fitted k in all flexible ones are smaller, suggesting reduced coal-CH₄ affinity. This coincides with our findings in Section 3.3 regarding the isosteric heat of adsorption.

Fig. 2(b) presents the variation of volumetric strain ($\varepsilon_V = (\langle V \rangle - V_0)/V_0$, where $\langle V \rangle$ denotes the ensemble averaged matrix volume and V_0 is the initial matrix volume with zero CH₄ loading) with P , along with the Tóth model fitting curves, where $n(P)$ is replaced by ε_V in Eq. (1). Due to the significant volume fluctuations (see Fig. S6) in amorphous anisotropic matrices during pressure control, particularly compared to the relatively stable number of adsorbed CH₄ molecules, the error bars for ε_V are larger, especially at high P . Nevertheless, the Tóth equation effectively captures the relationship between ε_V and P , with the maximum ε_V reaching 3.6 %~5.7 % at 500 bar.

As shown in Fig. 3, CH₄ loading and ε_V have a good linear correlation, in line with the simulation work by Potier et al. [38] on microporous carbon and experimental work by Bustin et al. [16] and Tang et al. [77] on coal samples. This suggests that the increased porosity due to swelling can accommodate more CH₄ molecules. Our samples exhibit distinct initial pore structures (see Table 2), leading to varying deformation responses to CH₄ adsorption, as indicated by the slopes of the fitting equations in Fig. 3. Notably, Sample #4, with the largest initial pore volume, exhibits the smallest swelling at the same CH₄ loading, while Sample #2, with the smallest initial pore volume, shows the largest ε_V . This may be due to higher local pressures and stresses in smaller pores, resulting in greater deformation.

3.2. Pore structure evolution upon CH₄ adsorption

To quantitatively characterize micropore property evolution induced by CH₄ adsorption, we calculate the occupiable pore volume (using CH₄ and He as probes), porosity (the ratio of occupiable pore volume to matrix volume), PSD, and average pore size for five samples at various equilibrium P . We use Sample #3 results as a representative, with additional data for the other two samples provided in SI. To ensure accurate sampling of atomic coordinates for pore structure analysis, we select five distinct frames whose volumes closely match the average matrix volume over the equilibrium run. This approach minimizes the impact of volume fluctuations and ensures that the pore volume calculations correspond to the equilibrium state of matrix.

Fig. 4 shows the variations of CH₄-occupiable pore volume (V_{P_o}) and porosity (φ_{P_o}) with adsorption pressure. At $P = 0$, V_{P_o} and φ_{P_o} are 0.0632 $\pm$ 0.0013 cm³/g and 7.6108 $\pm$ 0.1215 %, respectively; at $P = 500$ bar, V_{P_o} and φ_{P_o} are 0.1188 $\pm$ 0.00356 cm³/g and 13.4816 $\pm$ 0.3980 %, respectively, representing 87.97 % and 77.14 % increases, respectively. The less significant increase in φ_{P_o} relative to V_{P_o} is because φ_{P_o} is dependent on V_{P_o} and matrix volume, while both of them increase with P .

To further explore how micropores evolve during matrix deformation, we plot the PSDs at five representative pressures (0, 5, 50, 200, and 500 bar) in Fig. 5(a). The most significant change is the formation of additional pores of the same size (3.3 ~ 5.2 Å). For larger pores (9.1 ~ 9.6 Å), there is a slight shift to larger sizes and a decrease in peak intensity, indicating expansion and reduced frequency of these pores. To grasp the overall pore size variations, we calculate the average pore size using a weighted average approach, where each pore size is multiplied by its frequency (or probability), with the sum of these products divided by the total frequency, as shown in Fig. 5(b). Although the average pore sizes at most P are slightly smaller than the initial one, the overall trend remains largely intact. This supports the observation from the PSDs that the adsorption-induced increase in porosity is primarily due to the creation of additional pores of existing sizes, combined with minor expansion and reduced frequency of larger pores. Similar observations are shown in Fig. S8 for two other samples.

To intuitively and qualitatively observe the evolution of microporous space within coal matrix, we perform additional NVT simulations on

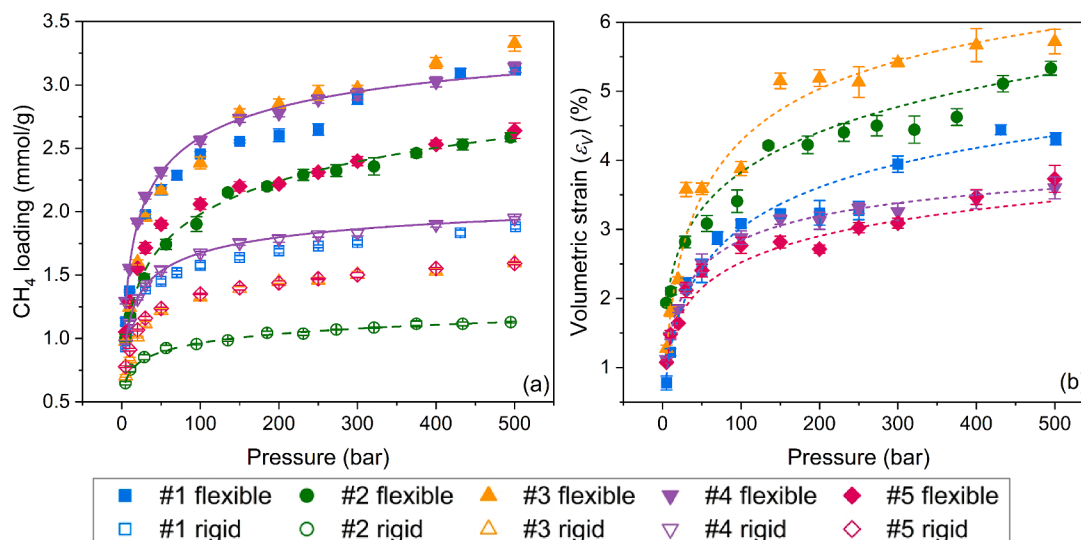


Fig. 2. (a) CH₄ adsorption isotherms (CH₄ loading denotes the total adsorption amount); (b) volumetric strain of the five samples. All the lines represent the fitted curves using the Toth equation.

Table 3

Fitted parameters of the Toth isothermal model.

	$n_{\max}(\text{mmol/g})$		$k(\text{bar}^{-1})$		t	
	Flexible	Rigid	Flexible	Rigid	Flexible	Rigid
Sample #1	5.046	2.331	2.102	5.645	0.277	0.320
Sample #2	3.418	1.281	1.769	9.626	0.323	0.346
Sample #3	4.689	2.005	0.380	1.910	0.368	0.354
Sample #4	3.796	2.358	0.831	2.948	0.404	0.358
Sample #5	3.329	1.988	1.788	4.714	0.338	0.326

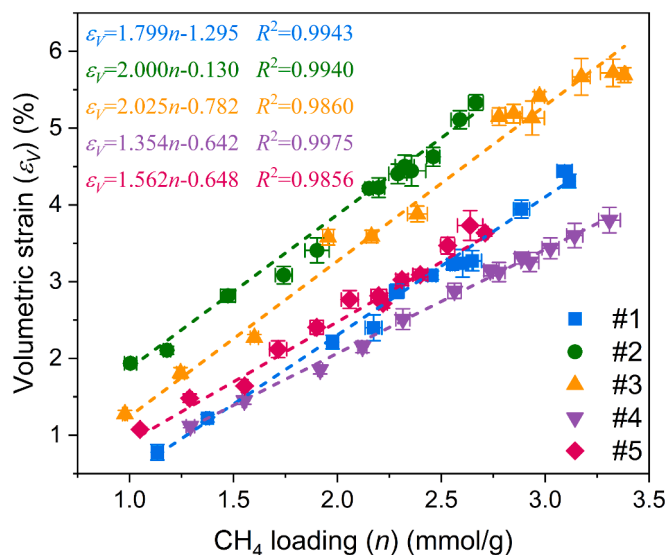


Fig. 3. Relationship between ε_v and CH₄ loading. The dashed lines represent the linear fitting.

CH₄ for both flexible and rigid cases (each lasting 4 ns: 2 ns for equilibrium and 2 ns for production) following the equilibration of previous hybrid GCMC/MD simulations. We divide the matrix into approximately 80,000 2D spatial bins and calculate the 2D CH₄ number density as shown in Fig. 6. The non-blue regions indicate the areas where CH₄ molecules have existed within the matrix. For the rigid matrix, CH₄ occupancy increases at low pressures (5 and 50 bar) but stabilizes after 50 bar, indicating adsorption saturation, in line with the adsorption

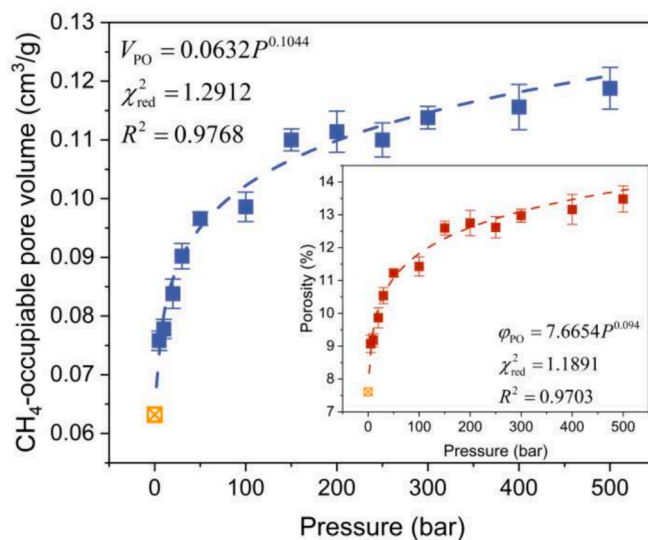


Fig. 4. Evolution of the CH₄-occupiable pore volume with adsorption pressure. Inset is the evolution of porosity. Orange squares indicate the initial pore volume and porosity with zero CH₄ loading. Similar observations are shown in Figure S7 for other two samples. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

isotherms. In contrast, the flexible case shows dynamic changes in the pore space. The adsorption-induced deformation allows CH₄ molecules to access more space within the coal matrix through macromolecular rearrangement and swelling. The flexible matrix creates additional pores, accommodating more CH₄ molecules, which corresponds to the increased PSDs shown in Fig. 5(a). Quantitatively, dividing the number of bins with non-zero density values by the total bin numbers, we obtain a ratio representing the pore frequency in a specific dimension. In Fig. 6, these ratios remain relatively stable for rigid matrices (20.4 %, 21.5 %, 21.8 %, 21.9 %) but continuously increase for flexible matrices (24.1 %, 34.1 %, 36.5 %, 39.0 %) as P increases. Similar trends are observed for other 2D density profiles (XZ-plane and YZ-plane), as shown in Fig. S9. Additionally, because more smaller pores are filled with CH₄ molecules, the average CH₄ densities in the flexible matrices are slightly higher than in the rigid ones, as depicted in Fig. 7 and Fig. S10 for other samples.

Fig. 8 reveals an intriguing observation: the incremental changes in

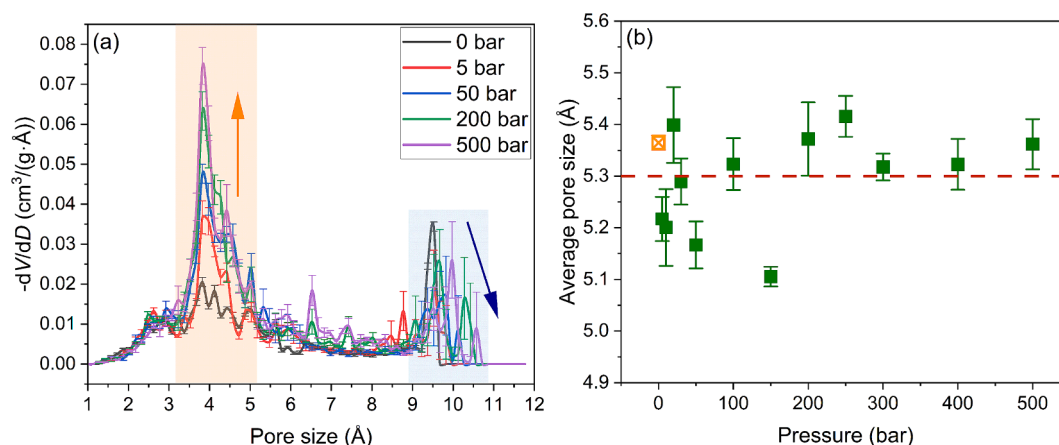


Fig. 5. Evolution of (a) PSDs; (b) average pore size with pressure. (a) 0 bar denotes the original coal matrix PSD with zero CH₄ loading. Two arrows indicate the PSD variation trends with pressure. (b) Red dashed line indicates the overall average pore size and the orange block denotes the initial average pore size. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

matrix volume and pore volume (using CH₄ and He as probes) at various P exhibit a strong linear correlation. Notably, the increase in the CH₄ probe-occupiable pore volume closely matches the matrix volume increase, indicating that the volumetric expansion of coal matrix is mainly driven by the increase in CH₄ probe-occupiable pores. Additionally, matrix deformation significantly increases the space available for He, suggesting the formation of more ultra-micropores (within the 2.58 to 3.73 Å range) due to adsorption-induced deformation. Similar phenomena are also observed for other samples (Samples #4 and #5) as shown in Fig. S11.

Besides the pore structure alteration induced by CH₄ adsorption, we present the matrix structural properties by computing skeletal density using CH₄ and He as probes in Fig. 9(a). The gradual decrease in bulk coal matrix density is expected, as the matrix bulk volume increases with P . Consistent with Fig. 8, the matrix volume—excluding the pore volume accessible to CH₄—remains essentially constant, resulting in relatively stable matrix skeletal density. However, as matrix deformation significantly enhances the available space for He compared to CH₄, an increase in the skeletal coal matrix density (He as probe) with pressure is observed, which follows a power function. This increase suggests compaction or structural rearrangement of the solid framework on a larger scale.

Inspired by this observation, we calculate the radial density distribution (RDD) for all carbon atoms to explore the carbon skeletal structure differences between flexible and rigid matrices under CH₄ adsorption, as shown in Fig. 9(b). We choose RDD over radial distribution function (RDF) because the former provides an absolute number density, unaffected by any changes in matrix volume during deformation, making it more reliable for comparing different solid configurations. Peaks #1 to #4 represent C–C bond lengths in aromatic and saturated structures (#1: 1.35 Å, #2: 1.55 Å), carbons separated by two and three bonds in aromatic structures (#3: 2.4 Å, 2.8 Å), and average spacing between stacked polyaromatic units (#4: 3.6 Å), respectively [50]. These peak intensities remain relatively constant, indicating that the matrix solid structure does not undergo substantial changes with CH₄ adsorption. The matrix retains its structural integrity without significant rearrangement or densification of C–C atoms, consistent with the hybrid GCMC/MD algorithm ensuring zero effective stress on solid structures. Thus, the increase in skeletal density with P alongside the stable local density from RDD suggests the macroscopic compaction of coal matrix without significant alteration to the local atomic environments.

3.3. Adsorption behavior variations upon matrix deformation

The isosteric heat of adsorption (q^{st}), which quantifies the binding affinity between adsorbate and adsorbent, is investigated for both rigid and flexible matrices. From a statistical mechanics perspective, q^{st} correlates with the fluctuations in the configurational (potential) energy U of the system and the adsorbed quantity N within the GCMC ensemble [78,79]:

$$q^{\text{st}} = k_{\text{B}}T - \frac{\partial \langle U \rangle}{\partial \langle N \rangle} = k_{\text{B}}T - \frac{\langle UN \rangle - \langle U \rangle \langle N \rangle}{\langle N^2 \rangle - \langle N \rangle^2} \quad (2)$$

where k_{B} is the Boltzmann constant. For flexible matrix, U includes all potential energy terms: van der Waals and Coulombic pairwise energy, long-range K-space energy and bonded molecular energy within the coal matrix. In contrast, for rigid matrices, only the van der Waals pairwise energy (solid–gas and gas–gas interactions) varies, as all solid atoms are immobile. Note that we employ the fluctuation method to calculate q^{st} due to its advantages in molecular simulations, particularly for flexible matrices. Unlike the Clausius-Clapeyron equation [80], which requires data from multiple temperatures, the fluctuation method allows for accurate and direct calculation from simulation data at a single temperature. Additionally, it captures microscopic fluctuations and interactions within the flexible matrix, offering a more detailed representation of the adsorption process in non-ideal and complex systems.

Fig. 10 displays the variation of q^{st} with respect to CH₄ loading and P . The simulated data show larger variations in q^{st} for flexible matrices, as indicated by error bars, which is due to the significant influences of solid atom movements on the total potential energy fluctuations. The magnitude and trend of q^{st} agree well with experimental and other simulation results [81–84]. As shown in Fig. 10(a), q^{st} generally decreases as CH₄ loading increases. The flexible matrices accommodate more CH₄ molecules than the rigid ones at lower q^{st} . Specifically, CH₄ adsorption in the flexible matrices releases lower heat (approx. 14 ~ 19 kJ/mol, blue oval area) compared to the rigid matrices (approx. 18 ~ 22 kJ/mol, orange oval area) at higher pressures (>100 bar). This implies that less energy is required to desorb CH₄ from the adsorbed state to the bulk state when P drops, thus facilitating gas desorption. Fig. 10(b) shows that at low P , the rapid decrease in q^{st} reflects the initial occupation of the most favorable adsorption sites, followed by a plateau as adsorption becomes less exothermic with increased pressure and the occupation of less favorable sites, marking the completion of the pore-filling process.

As mentioned above, q^{st} in flexible matrices accounts for all potential

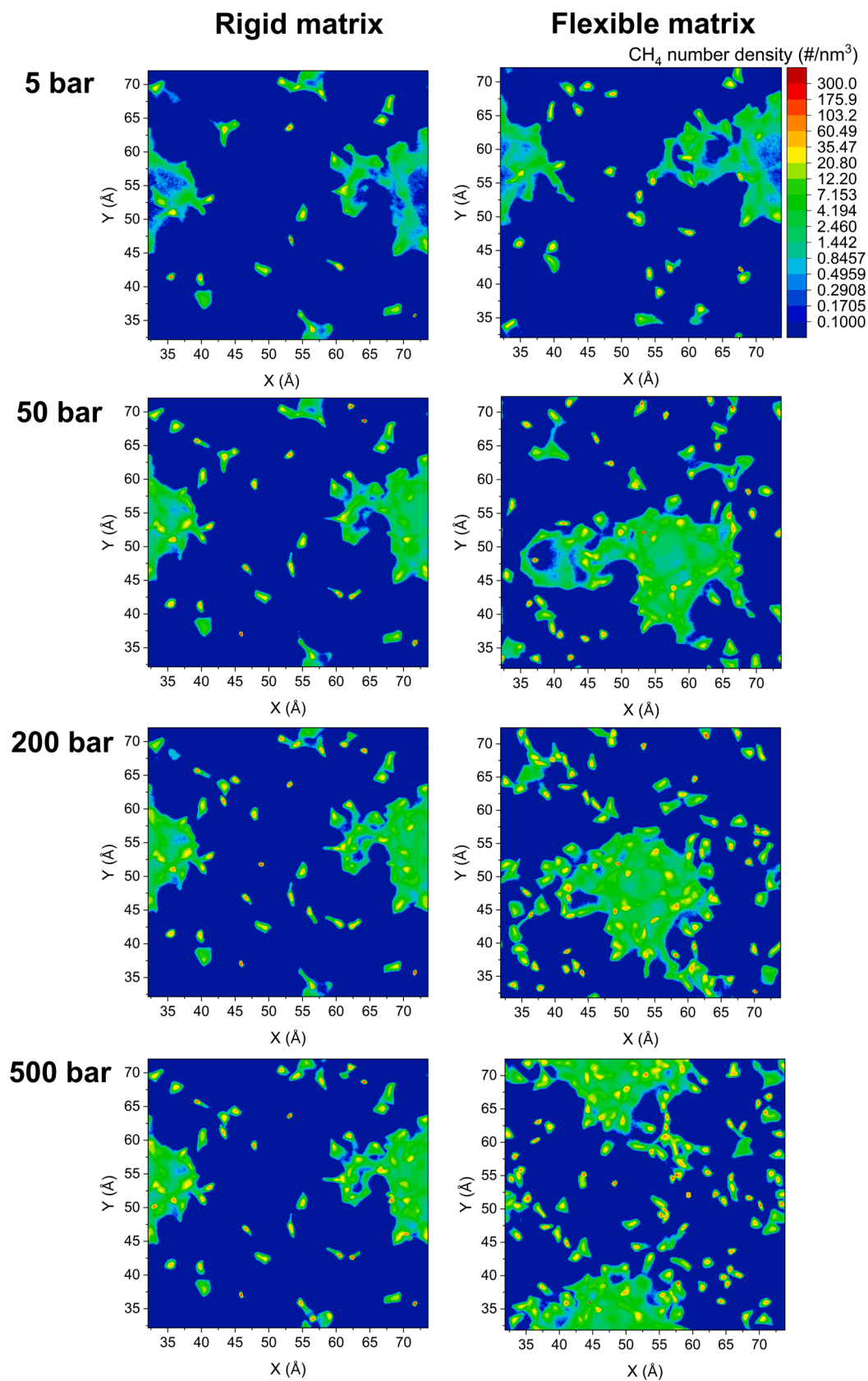


Fig. 6. Comparison of CH₄ 2D number density profiles in rigid and flexible coal matrices. The number density is projected in the Z-direction onto the XY-plane.

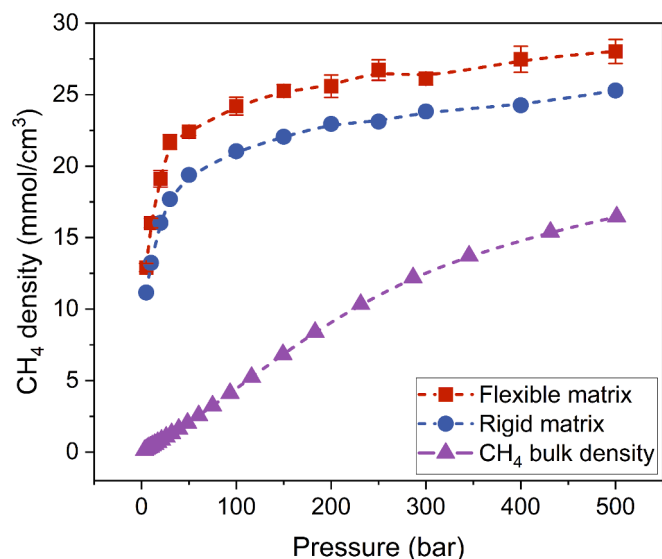


Fig. 7. Comparison of average CH₄ densities in flexible and rigid matrices. CH₄ bulk density is obtained from GCMC simulations. CH₄ densities are calculated by dividing the CH₄ loading amount (as shown in Fig. 2(a)) by CH₄ occupiable volume (as shown in Fig. 4) at various equilibrium pressures. Dashed lines are included for eye guidance and enhanced visualization.

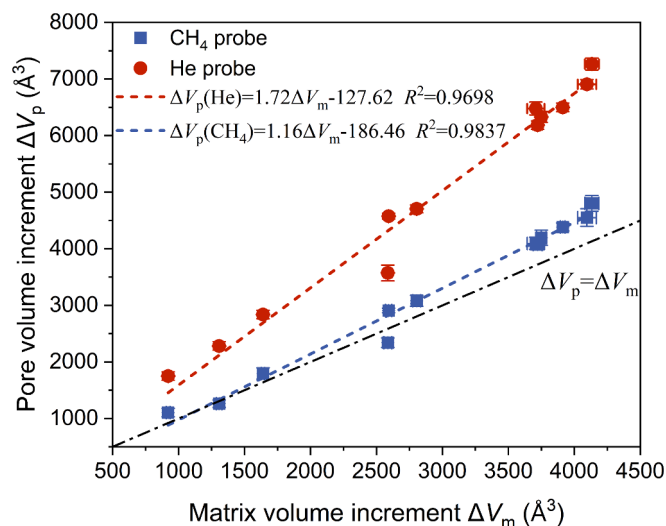


Fig. 8. Relationship between pore volume and matrix volume increment.

energy changes, including those due to the structural flexibility. To isolate the contribution of solid–gas and gas–gas interactions to q^{st} , we calculate the total energy between solid atoms and CH₄ molecules, as well as the energy between CH₄ molecules themselves, as shown in Fig. 11.

First of all, it is evident that solid–gas interaction is more dominant than gas–gas interaction in determining q^{st} . At low P , CH₄ molecules are sparsely distributed. However, as P increases, CH₄ intermolecular interactions gradually dominate, which increases q^{st} . At higher P , the matrix becomes saturated with CH₄ and q^{st} reaches a plateau for both rigid and flexible matrices. For solid–gas interactions, as the flexible matrix deforms, additional favorable adsorption sites (pores) are generated, thereby increasing q^{st} compared to the rigid matrix. Comparing Figs. 10 and 11, q^{st} for flexible matrix is larger than that for the rigid matrix when only considering solid–gas interaction, but when the total potential energy is involved, the flexible matrix shows a smaller q^{st} . In the rigid matrix, all the heat released during adsorption directly

contributes to the change in the van der Waals potential energy of the system. In contrast, the total potential energy in the flexible matrix includes not only the van der Waals interactions between the gas molecules and the matrix but also the internal energy associated with the matrix's structural changes. During CH₄ adsorption, the matrix can deform to accommodate gas molecules, converting some of the adsorption energy into other free energy (such as entropy), leading to a lower apparent q^{st} . Thermodynamically, the flexible matrix introduces additional degrees of freedom through deformation, acting as an energy sink that absorbs some of the energy that would otherwise increase the isosteric heat. Consequently, with lower heat release (less binding energy), it is reasonable to hypothesize that CH₄ desorption in flexible matrices might be easier compared to that in rigid matrices.

4. Conclusions

Using hybrid GCMC/MD approach, in this work, we investigate the two-way coupling mechanism of adsorption-induced deformation in five independent microporous coal matrices. We find that the Tóth isothermal model can effectively describe CH₄ adsorption behaviors in both flexible and rigid matrices, as well as volumetric strain isotherms. In flexible matrices, pore volume and porosity increase mainly by generating additional pores of similar sizes (3.3 ~ 5.2 Å) to the original pores without CH₄ loading, thus maintaining a relatively constant average pore size across different pressures. The observed linear relationship between volumetric strain and CH₄ loading suggests predictable matrix behaviors under varying conditions. Matrix swelling matches the expansion of CH₄-occupiable pore volume, but is less than the expansion of He-occupiable pore volume, resulting in increased skeletal density when using He as a probe. Additionally, the local radial density distribution of C–C atoms indicates that the immediate environment around carbon atoms remains relatively unchanged, preserving short-range structural order even as the matrix becomes denser overall. Furthermore, CH₄ adsorption behaviors in a flexible matrix are significantly influenced by pore structural changes, impacting both adsorption capacity and the interaction energy between CH₄ molecules and solid atoms. In a flexible matrix, a portion of the energy released during adsorption is absorbed by the matrix to accommodate its deformation, rather than being entirely released as heat, as in a rigid matrix. Such energy absorption results in a lower net energy change, measured as the isosteric heat of adsorption. Consequently, gas desorption in a flexible matrix may require less energy compared to a rigid matrix.

The hybrid GCMC/MD algorithm assumes that the imaginary reservoir pressure is equal to that in NPT ensemble. This setup leads to the observation of the maximum swelling of the matrix as pointed out by other studies [43,44]. However, *in-situ* conditions require the consideration of overburden compressive stress, complicating the simulation scheme for the adsorption-deformation coupling. Future research should focus on gas adsorption under *in-situ* reservoir conditions to provide more accurate representations of adsorption mechanisms. Moreover, future research should also explore the diffusion behavior in nanoporous coals, particularly how matrix deformation influences gas transport efficiency into larger pores and fractures. Understanding this coupled effect is crucial for optimizing gas recovery processes, as deformation-induced changes in pore structure could significantly impact diffusion pathways and the overall effectiveness of gas transport.

CCRediT authorship contribution statement

Quanlin Yang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Junhua Xue:** Supervision, Methodology, Conceptualization. **Haifei Lin:** Supervision, Methodology, Conceptualization. **Zhehui Jin:** Writing – review & editing, Supervision, Resources, Investigation, Funding acquisition, Formal analysis, Conceptualization.

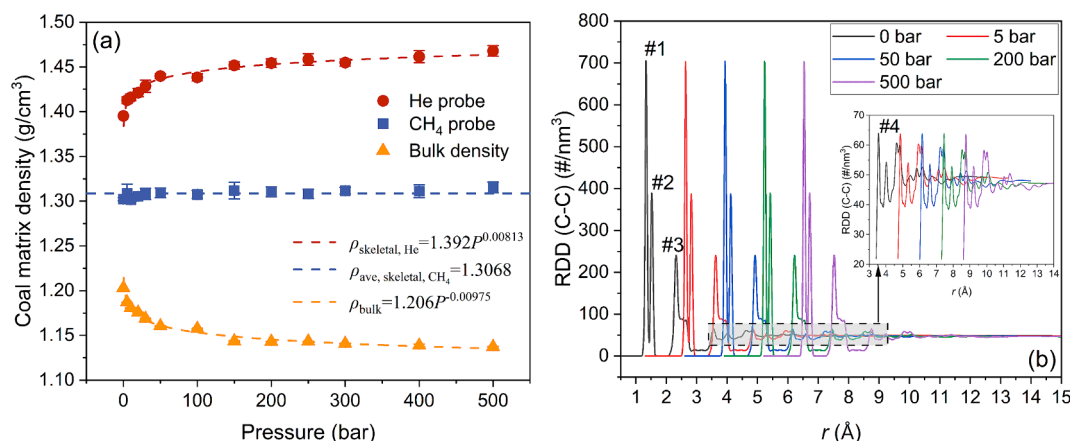


Fig. 9. (a) Coal matrix density and (b) radial density distribution of carbon atoms. For clarity, curves have been shifted along r -axis for 5 bar (+1.3 Å), 50 bar (+2.6 Å), 200 bar (+3.9 Å) and 500 bar (+5.2 Å). The same phenomena are observed for Sample #5 as shown in Fig. S12.

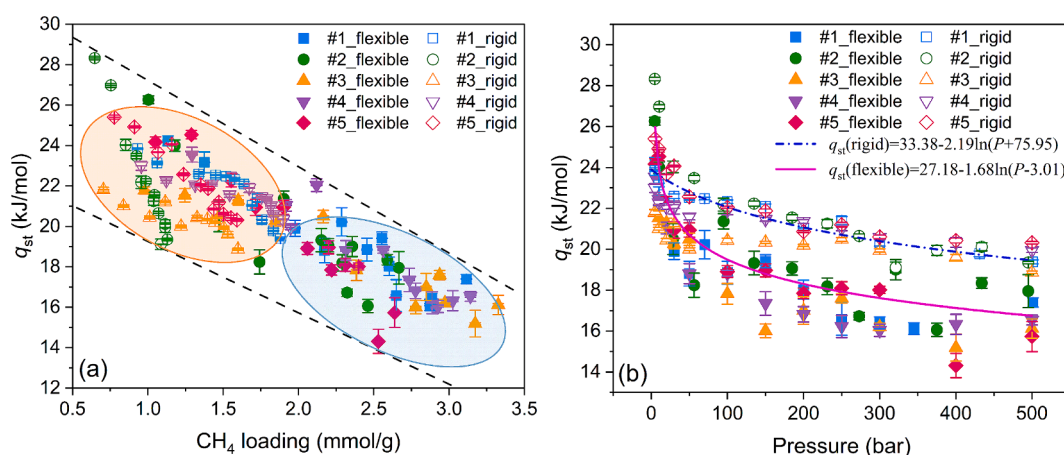


Fig. 10. Isosteric heat of adsorption of CH₄ in rigid and flexible matrices versus (a) CH₄ loading; (b) pressure.

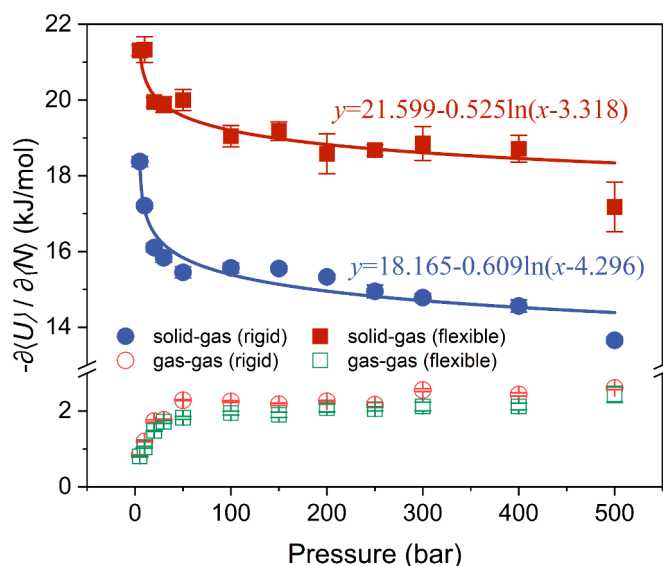


Fig. 11. Different contributions to isosteric heat of adsorption in flexible and rigid matrices. Similar trends for other samples are observed in Fig. S13.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2024.155321>.

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